

Extracting the Group Index of Silicon Waveguides Using Unbalanced Mach-Zehnder Interferometer Designs

Katie Bian

Abstract—The low loss of silicon waveguides is a critical factor in the success of silicon photonic chips, and is largely characterized by the group index. To minimize device errors and verify expected performance post-fabrication, it is important to use simulations to predict expected behavior as well as perform empirical measurements to confirm the predictions. This report uses various unbalanced Mach-Zehnder Interferometer designs with changes in path length difference to extract the Free Spectral Range and group index of 500nm wide and 220nm thick silicon strip waveguides at 1550nm in the fundamental TE mode. Results show that the Free Spectral Range values were closely aligned to simulation results, whereas group index values were more susceptible to process perturbations, such as noise and unaccounted for variations during fabrication. Finally, the relationship between Free Spectral Range and path length difference is investigated and experimental data strongly matches mathematical equations.

I. INTRODUCTION

WITH the use of AI and large-scale cloud computing becoming more ubiquitous in recent years, data centers are met with an unprecedented demand for increased bandwidth density in order to support the high-volume computation that these new technologies rely on. Current electrical solutions involve adding more inter-chip channels to boost data rates, but these efforts are fundamentally limited by the available perimeter of a silicon die [1]. In light of this bandwidth ceiling, the idea of using silicon photonics to replace electrical interconnects with optical ones is gaining tremendous traction. Key benefits include the ability to achieve higher bandwidth density through 3D co-packaged optics, and compatibility with modern CMOS fabrication processes [2].

To achieve large-scale adoption, photonic chips are required to maintain or exceed strict signal integrity requirements [3]. One of the important features that enables this compliance is the ability of on-chip silicon waveguides to route light with high precision and low loss. The design of such a waveguide relies heavily on knowing its group index (n_g), for n_g not only characterizes the speed at which an optical pulse envelope will travel (known as the group velocity (v_g)) [4], but the wavelength-dependent nature of n_g also informs the amount of chromatic dispersion the signal will suffer from [5].

Having an accurate design can reduce the need for repeated fabrication runs, which are both time-consuming and costly. To minimize excessive design cycles due to incorrect device behavior, n_g is often first obtained through simulations as a reasonable prediction of what might be expected from the physical product. However, the statistical variations and process limitations associated with fabrication oftentimes introduce deviations from the simulated results. It is therefore crucial to also measure the experimental n_g of the fabricated

device and ensure that its value falls within an acceptable range of the theoretical calculations. In this regard, a common method of extracting n_g is to use a Mach-Zehnder Interferometer (MZI) and the principle of interference to produce a series of peaks and troughs in the transmission spectrum, from which n_g can be calculated by measuring the Free Spectral Range (FSR) [6]. Apart from the aforementioned utilization, the MZI configuration can be found in a myriad of applications, ranging from complex optical switching, to the modulation of light in Wavelength Division Multiplexing (WDM) systems [1][7].

This report uses multiple unbalanced MZIs with varying path length differences (ΔL s) to experimentally extract the n_g of 500nm wide by 220nm thick silicon strip waveguides. The different MZI designs are also used to investigate the relationship between FSR and ΔL . The devices were fabricated on 200mm diameter Silicon On Insulator (SOI) wafers using Applied Nanotools Inc.'s NanoSOI MPW process, based on electron beam lithography [8]. Finally, measurements were performed with 1550nm TE polarized light at 25°C.

II. THEORY

A cross-sectional image of a strip waveguide on an SOI wafer is shown in Figure 1. The higher index of refraction silicon core (n_{core}), surrounded by a lower index cladding (n_{clad}) allows light to propagate down the waveguide by way of Total Internal Reflection (TIR) [9]. More specifically, the *apparent* refractive index that the light sees as it travels within the waveguide is a combined contribution from both n_{core} and n_{clad} , and is known as the effective index of refraction (n_{eff}) [2].

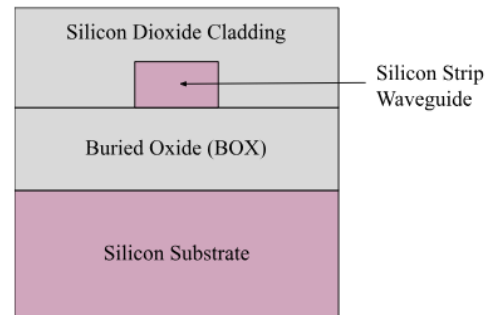


Fig. 1: Cross section of a silicon strip waveguide core surrounded by silicon dioxide cladding, on an SOI wafer. With reference to [2].

n_{eff} is an indispensable quantity for describing the plane wave characteristics of light propagation in waveguides. The electric field component of a generic optical plane wave has the equation [2]:

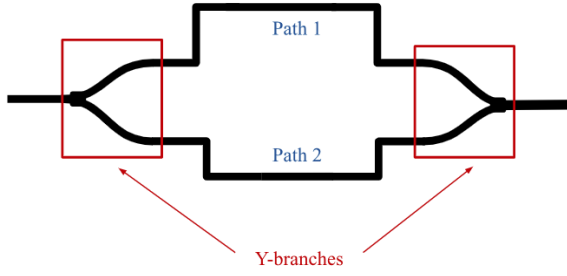


Fig. 2: Example unbalanced MZI configuration. In general, n_{eff} , waveguide length, and loss can be different between the two paths. With reference to [2].

$$E = E_0 e^{i(\omega t - \beta z)} \quad (1)$$

E_0 contains the amplitude and the complex exponential is the phase, where $\beta = \frac{2\pi n_{eff}}{\lambda}$. In the case of an optical beam traveling through an MZI, with a configuration such as the one in Figure 2, the light is first evenly split into two separate waveguide paths. Each path will have an intensity (I) equal to $\frac{I_{in}}{2}$ and an electric field (E) equal to $\frac{E_{in}}{\sqrt{2}}$. In the most general case, the n_{eff} of the two waveguides are not equal, and propagation of the light will introduce a phase shift that is related to the path lengths (L) and the power loss (α) [2]:

$$E_{path1} = \frac{E_{in}}{2} e^{-i\beta_1 L_1 - \frac{\alpha_1}{2} L_1} \quad (2a)$$

$$E_{path2} = \frac{E_{in}}{2} e^{-i\beta_2 L_2 - \frac{\alpha_2}{2} L_2} \quad (2b)$$

The paths are recombined on the other side with a second Y-branch according to the equation [2]:

$$\begin{aligned} E_{out} &= \frac{E_{path1} + E_{path2}}{\sqrt{2}} \\ &= \frac{E_{in}}{2} (e^{-i\beta_1 L_1 - \frac{\alpha_1}{2} L_1} + e^{-i\beta_2 L_2 - \frac{\alpha_2}{2} L_2}) \end{aligned} \quad (3)$$

Finally, in terms of intensity, equation (3) can be written as:

$$I_{out} = \frac{I_{in}}{4} |e^{-i\beta_1 L_1 - \frac{\alpha_1}{2} L_1} + e^{-i\beta_2 L_2 - \frac{\alpha_2}{2} L_2}|^2 \quad (4)$$

Depending on the phase shifts applied to each of the MZI arms, $\frac{I_{out}}{I_{in}}$ can theoretically be a maximum of 1 and a minimum of 0, representing total constructive and destructive interference at the MZI output, respectively. This is visually observed by the peaks and dips in the gain spectrum, as highlighted in Figure 3. The FSR can be determined by measuring the period of the plot, from which n_g can be extracted according to the following equation [6]:

$$FSR = \frac{\lambda^2}{n_g \Delta L} \quad (5)$$

where ΔL is the path length difference between the two arms of the interferometer.

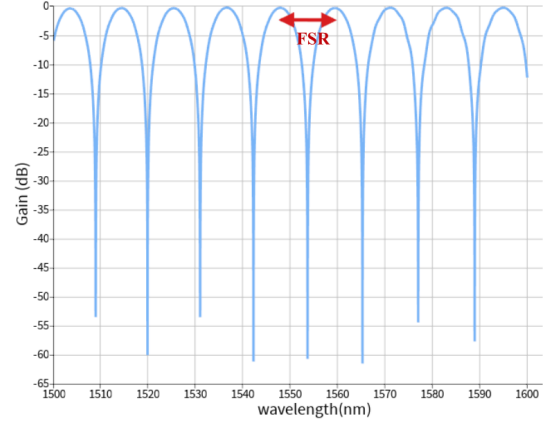


Fig. 3: Example gain spectrum of an unbalanced MZI with no insertion loss, simulated using Lumerical INTERCONNECT.

III. MODELING AND SIMULATION

The 500nm wide and 220nm thick silicon strip waveguides used in this report were simulated using Lumerical MODE's Finite Difference Eigenmode (FDE) solver to generate the mode profiles of the fundamental TE mode, as shown in Figures 4(a)-(c). To ensure an accurate simulation, the FDE boundaries were adjusted until the electric field intensity decayed to less than $1e-6$ at the perimeter, and the number of mesh cells in both dimensions of the solver was set to 100. The energy density plot in Figure 4(c) depicts the ability of the waveguide to contain almost all the light inside the core, thus allowing for routing with low loss.

Performing a wavelength sweep between 1500nm and 1600nm of the fundamental TE mode allowed the relationship between n_{eff} and wavelength (λ), and n_g and λ to be plotted, as done in Figures 4(d)-(e). As is evident, both n_{eff} and n_g are not constant in λ . In fact, n_g can be calculated from n_{eff} through the following equation:

$$n_g = n_{eff} - \lambda \frac{dn_{eff}}{d\lambda} \quad (6)$$

The dependence of n_{eff} on λ is an empirically measured quantity and can be modeled through various different fitting equations, one of the simplest being a Taylor expansion around a center wavelength (λ_0):

$$n_{eff} = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2 \quad (7)$$

The coefficients are in turn related to n_{eff} , n_g , and the dispersion coefficient (D) at the specific λ_0 wavelength :

$$n_{eff} = n_1 \quad (8a)$$

$$n_g = n_1 - n_2 \lambda_0 \quad (8b)$$

$$D = -2\lambda_0 \frac{n_3}{c} \quad (8c)$$

where c is the speed of light in vacuum.

To determine these above parameters, this report used the Fabry-Perot cavity resonance condition to find n_{eff} :

$$2n_{eff}dL = m\lambda_0 \quad (9)$$

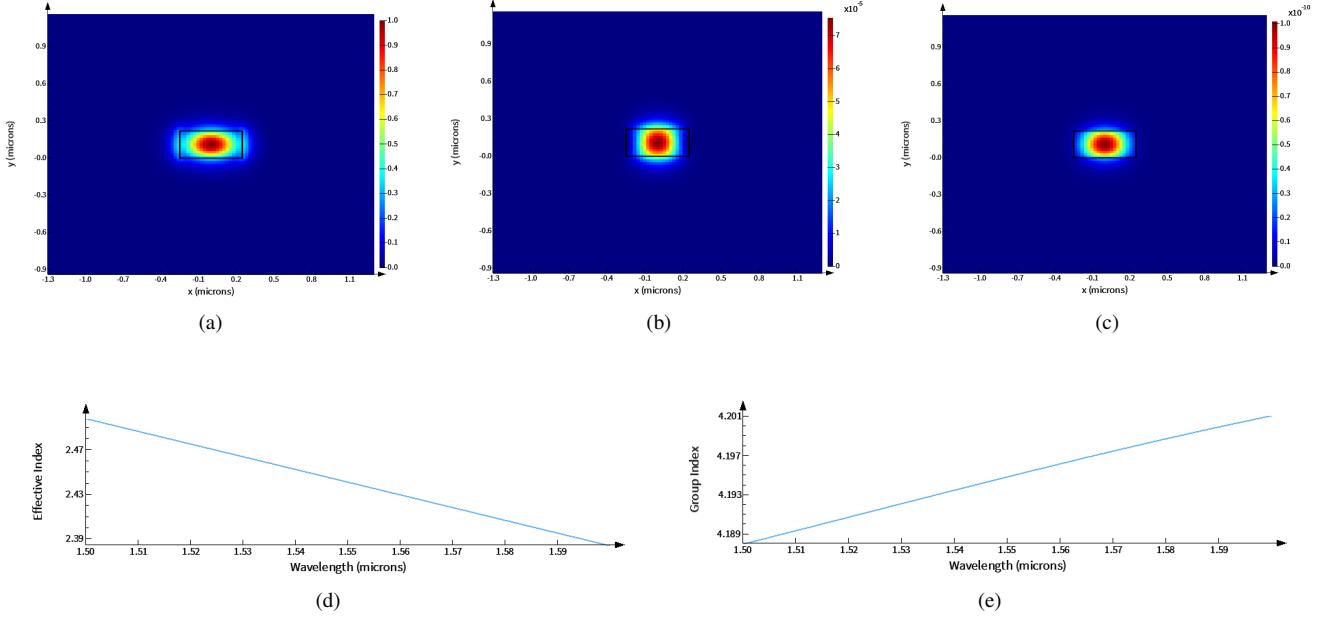


Fig. 4: Lumerical MODE simulations of a 500nm by 220nm silicon strip waveguide cross section for the fundamental TE mode, showing (a) the electric field intensity, (b) the magnetic field intensity, (c) the energy density, (d) the plot of n_{eff} vs. wavelength, and (e) the plot of n_g vs. wavelength.

where m is the mode number. Next, n_g was obtained from a simple derivation of equation (5), using the extracted FSR from experimental data:

$$n_g = \frac{\lambda^2}{(FSR)\Delta L} \quad (10)$$

Finally, D was calculated from curve fitting using Matlab's *lsqcurvefit* function to find n_3 .

Overall, the fundamental asset that allows for these calculations to be accomplished is the data collected from the MZIs. The iterations on MZI design for this report can be found in Table 1, along with the predicted FSR at 1550 nm for each. Identical waveguides are used in the upper and lower arms to eliminate the complexity in equation (4) associated with two different refractive indices and thus two different β s. Furthermore, if the loss in the waveguides are assumed negligible, equation (4) simplifies to:

$$\frac{I_{out}}{I_{in}} = \frac{1}{2}[1 + \cos(\beta\Delta L)] \quad (11)$$

This is the transfer function of an unbalanced MZI with negligible loss and identical waveguides.

TABLE I: MZI Design Iterations and the Theoretical FSR for Each

Design	L_1 (μm)	L_2 (μm)	ΔL (μm)	Theo. FSR (nm)
MZI1	63.58	88.98	25.40	22.52
MZI1p1	63.58	88.98	25.40	22.52
MZI1p2	63.58	88.98	25.40	22.52
MZI2	63.58	89.63	26.05	21.96
MZI3	63.58	93.98	30.40	18.82
MZI4	63.58	98.98	35.40	16.20
MZI5	63.58	103.98	40.40	14.19
MZI6	63.58	196.98	133.40	4.29
MZI7	63.58	108.98	45.40	12.63

As proof of concept, a simulation run was undertaken using an MZI circuit built in Lumerical INTERCONNECT and shown in Figure 5(a). The Y-branches were represented with an S-parameter file and a Lumerical MODE model was loaded in for the waveguides. Taking MZI6 as an example calculation for n_g , using equation (10), the transfer function is plotted in Figure 5(b), and the FSR at 1550nm extracted from INTERCONNECT is 4.29nm (see also Table 1). Therefore, the theoretical n_g for this design is:

$$n_g = \frac{1550nm^2}{(4.29nm)(133400nm)} = 4.20 \quad (12)$$

A similar approach was followed for analyzing the experimental data in the following sections, where the FSR was calculated by using a *findpeaks* function in Matlab and taking the difference between adjacent x-axis values.

To estimate the fabrication variations, the waveguide width was assumed to be between 510nm and 470nm, and the waveguide thickness between 223.1nm and 215.3nm. Figure 6 plots the worst case corners of n_g that were calculated in Lumerical MODE, along with the nominal design. The worst case corners were 510nmX215.3nm with an n_g of 4.17 and 470nmX223.1nm with an n_g of 4.26.

IV. FABRICATION

The MZIs were fabricated using Applied Nanotools Inc.'s 100keV electron beam lithography process. The SOI wafers were pre-diced into 25mmX25mm squares, before undergoing wafer cleaning for 15 minutes. Hydrogen silsesquioxane (HSQ) was used as the resist and the GDS MZI pattern was imprinted at the University of British Columbia (UBC) with a JEOL JBX-8100FS electron beam instrument. The developed

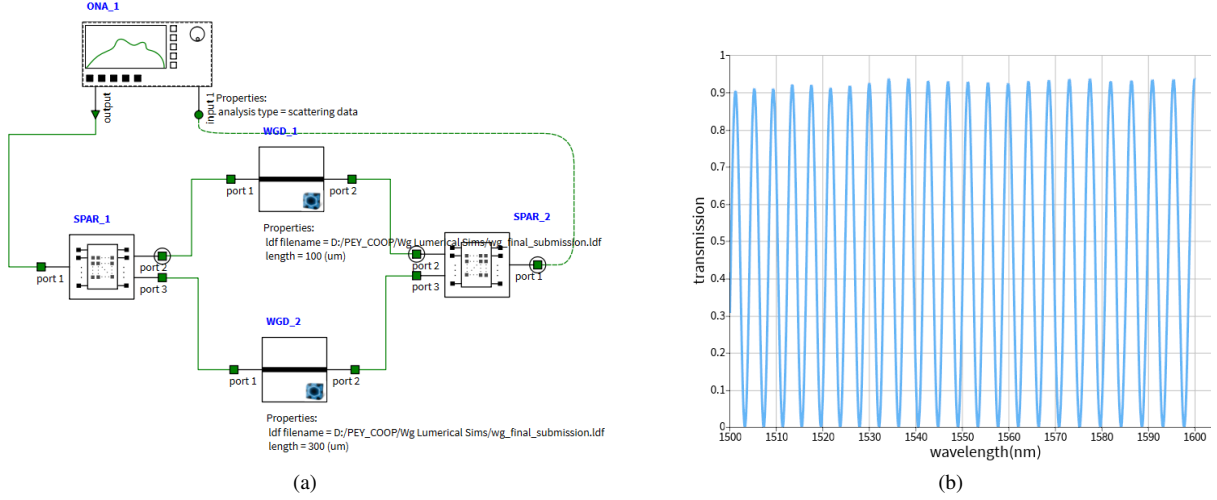


Fig. 5: Lumerical INTERCONNECT proof of concept simulation for MZI6 with (a) the circuit setup where Y-branches are represented through S parameter blocks and waveguide models are imported from Lumerical MODE, and (b) the resulting transmission spectrum.

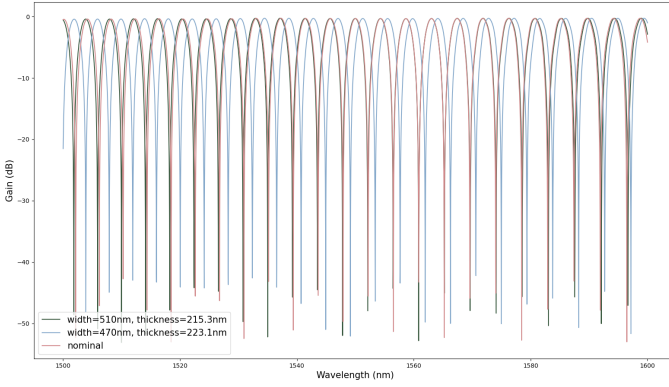


Fig. 6: Corner analysis plots for worst case corners and nominal design.

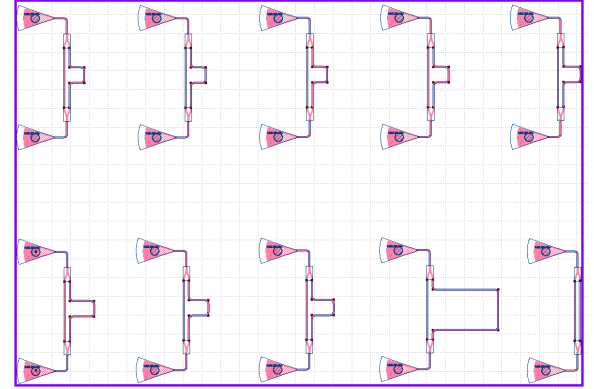


Fig. 7: KLayout GDS of the MZI iterations listed in Table 1, with an additional balanced MZI structure for calibration.

resists were then optically inspected for residues and defects, and subsequently etched with an anisotropic ICP-RIE process. The resist was removed with a 10:1 buffer oxide wet etch, and the device was inspected under a scanning tunneling microscope (SEM). Finally, $2.2\ \mu\text{m}$ thick oxide was deposited as the cladding through plasma-enhanced chemical vapor deposition (PECVD). A round of verification was performed before shipment. [8]

The GDS layout using KLayout is shown in Figure 7. Three MZI configurations were repeated to compare fabrication variability against the corner analysis results. Ideally, more than three devices should be used to properly characterize variability. However, due to area limitations, this could not be realized. One balanced MZI is included for calibration of the insertion loss due to the grating couplers.

V. EXPERIMENTAL DATA

The raw, uncalibrated MZI spectra in Figure 8(a) are plotted on the same graph as the simulated data from Lumerical INTERCONNECT. The simulations were obtained from the INTERCONNECT functionality built into the EBeam library in KLayout. In both the theoretical and experimental cases, the

insertion loss of the grating couplers is apparent from the finite bandwidth of the plots. However, the experimental data overall has much lower transmission compared to the theoretical, likely due to the simulation model not fully capturing all the sources of loss during measurement. Furthermore, there is a noticeable systematic shift towards shorter wavelengths in all the measured spectra, causing the center wavelength to no longer being at 1550nm. The implications of this result were noticeable when attempting to curve fit and interpolate FSR from the data, where the increased noise from the spectra edges had a larger impact on the calculations than if the data were centered around 1550nm. Despite this degradation in accuracy, reasonable FSR and n_g values were able to be extracted.

Before extraction, it is of interest to decouple the insertion loss of the grating couplers from the actual response of the MZI structure itself, so as to ensure the analysis of the data is not influenced by external factors. As such, a *polyfit* function was used in Matlab to fit a curve to the calibration structure's raw data spectrum, omitting the wavelength range near either ends due to noise, but ensuring 1550nm was

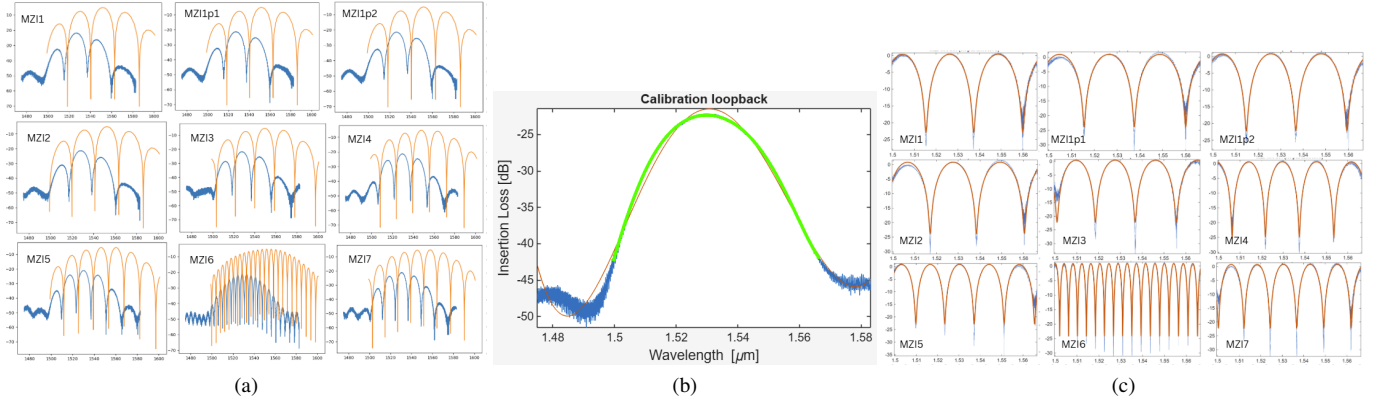


Fig. 8: Experimental data plots of the MZI designs listed in Table 1 depicting (a) raw data in blue overlaid with INTERCONNECT simulations in orange, (b) curve fit in green of the raw calibration structure data in blue, and (c) final MZI transfer function curve fit in red overlaid with de-embedded raw data in blue.

retained. Subsequently, Y-values from the fit were subtracted from the raw data to flatten out the transmission response, allowing for a closer resemblance to the simulated plot in Figure 3. To converge efficiently on a curve fitting solution that accurately models the transmission spectrum, Matlab code was written to analyze the data in a two-step process. Firstly, a *findpeaks* algorithm was used to calculate the FSR and in turn n_g at various λ . Once a preliminary set of n_g were found, they were then used as the starting point to deduce the coefficients needed for the n_{eff} Taylor expansion of equation (7). Finally, knowing n_{eff} , β was found to solve for the MZI transfer function. With an accurate transfer function fit, the n_{eff} was re-calculated with more refined values to ultimately produce an n_g vs. λ plot using equation (6).

The polynomial fit to the calibration structure and the de-embedded MZI spectra with the transfer function curve fit are displayed in Figures 8(b)-(c).

VI. ANALYSIS

Taking a more in-depth look into the data extraction of MZI6, the FSR values extracted from the *findpeaks* algorithm are plotted in Figure 9, from which the FSR for MZI6 at 1550nm is 4.192nm. Comparing this value against the theoretical FSR of 4.29nm in Table 1 results in a reasonable percent discrepancy of 2.28%. The final plot of n_g vs. λ , obtained by the two-step fitting process described in Section V, is shown in Figure 10. The experimental value of n_g extracted from this graph is listed in Table 2, along with the n_g and FSR for the other MZI designs. FSR values from experimental measurements aligned reasonably well with the theoretical values from Table 1. All MZI parameters in Table 2 were determined using the same process as was outline for MZI6.

To facilitate the analysis, Figure 11 contains all the experimental n_g vs. λ plots in one graph for easy comparison. There are a few unexpected negative sloped plots, which likely implies the presence of errors in the fitting that resulted in inaccurate extraction of the Taylor expansion coefficients, thus leading to a negative $\frac{dn_g}{d\lambda}$ value. This echoes the results in Table 2, as certain the experimental n_g values lie outside the corner analysis range on page 3. Although the designs

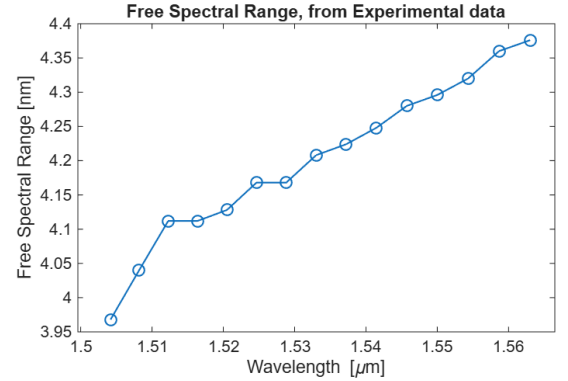


Fig. 9: FSR vs. Wavelength plot for MZI6, obtained from the *findpeaks* Matlab function.

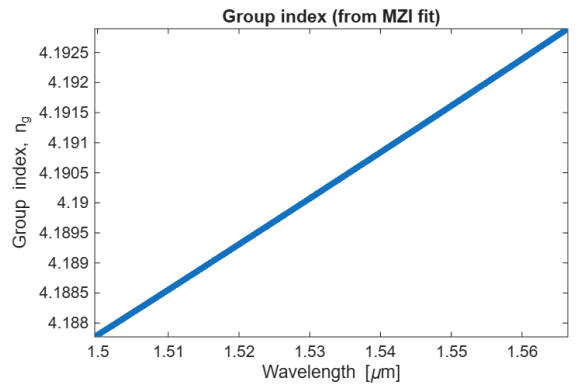


Fig. 10: n_g vs. Wavelength plot for MZI6, obtained from a two-step curve fitting process.

with n_g values outside the corner analysis do not necessarily align with the decreasing n_g vs. λ plots, the presence of these unpredicted results indicate the presence of sources of error that have caused the variability in the final outcome. These sources of error likely arise from the imperfect fitting of the data due to noise.

To study the fabrication variations, the three repeated MZI designs are plotted together in Figure 12. With the limited data available, there is negligible variation in these three design

TABLE II: Experimental n_g and FSR at 1550nm for Each MZI Design

Design	n_g	FSR (nm)
MZI1	4.10	22.62
MZI1p1	4.15	22.49
MZI1p2	4.12	22.78
MZI2	4.13	22.06
MZI3	4.17	19.00
MZI4	4.23	16.01
MZI5	4.21	14.03
MZI6	4.19	4.28
MZI7	4.19	12.50

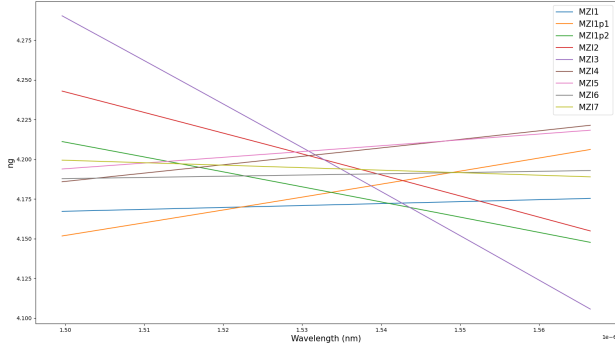


Fig. 11: n_g vs. Wavelength plots for all MZI designs listed in Table 1.

iterations, unlike what was predicted through corner analysis simulations in Figure 6, where shifts in the spectrum were more apparent. Due to the small sample size and the fact the three designs were placed close together in the layout (see Figure 7), the process variations can not be holistically determined. The only conclusion that can be made in this scenario is that the variations are negligible for these three specific iterations. In future fabrications, including more identical designs on the same wafer and across different wafers will yield a better picture of the true variation.

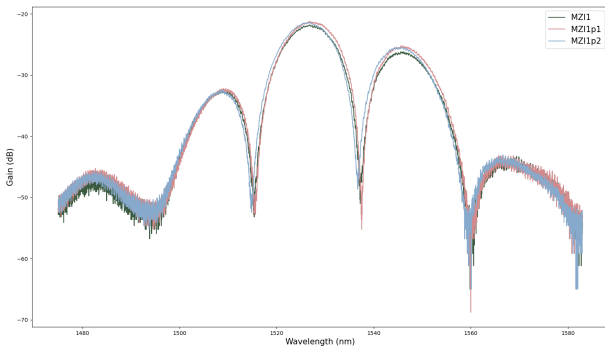


Fig. 12: Three transmission plots of identically designed MZIs to observe fabrication variations.

Finally, the experimental FSR from Table 2 was assessed against changes in ΔL in Figure 13 to investigate the relationship between the two parameters. The data points were fit in LoggerPro using an inverse curve of the form $\frac{A}{x}$, where A is a fitting parameter and x would be n_g in this case. The achieved R^2 was 0.9998, indicating that FSR is indeed inversely proportional to n_g , as expected from equation (5).

Future studies could look at collecting more data points at evenly spaced intervals, rather than the majority clustered together and a single data point farther away. This would provide more information to determine the fit parameter A with more accuracy.

VII. CONCLUSION

This report analyzed the transmission spectra of nine different unbalanced MZI designs, with identical β but varying ΔL . The mode profiles, n_g and n_{eff} dependence on λ , FSR, and corner analysis for the 500nm wide and 220nm thick silicon strip waveguide were simulated in Lumerical MODE and INTERCONNECT to obtain a theoretical prediction. The designs were fabricated using electron beam lithography by Applied Nanotools Inc., and measurements were performed via optical fiber interfacing with grating couplers. The resulting experimental FSR values aligned well with theoretical calculations, and the plot between FSR and ΔL showed a clear inverse relationship, as expected from the mathematical expressions. On the other hand, almost four out of nine n_g values showed unpredicted negative slopes in the n_g vs. λ graphs, along with a few experimental n_g values that fell outside the corner analysis range. This suggests sources of error, including noise in the data plots and unmodeled phenomena during fabrication, such as the systematic shift of the spectra toward shorter wavelengths.

REFERENCES

- [1] Y. Wan, W. He, *et al.*, "Integrating silicon photonics with complementary metal-oxide-semiconductor technologies," *Nat Rev Electr Eng*, 2025. DOI: <https://doi.org/10.1038/s44287-025-00223-0>.
- [2] L. Chrostowski, and M. Hochberg, *Silicon Photonics Design*. Cambridge: Cambridge University Press, 2015, ch.1, pp.3-22.
- [3] *Draft Standard for Ethernet Amendment 12: Media Access Control Parameters for 1.6 Tb/s and Physical Layers and Management Parameters for 200 Gb/s, 400 Gb/s, 800 Gb/s, and 1.6 Tb/s Operation*, IEEE P802.3dj/D2.1, Aug. 14, 2025.
- [4] J. Shen, *et al.*, "Integrated Pockels Modulators on Silicon Photonics Platform," *Adv. Phys. Res.*, vol. 4, no. 4, Apr. 2025. Accessed: Nov. 27, 2025. Available: <https://doi.org/10.1002/apxr.202400096>.
- [5] P. Hlubina, D. Ciprian, and M. Kadulova, "Measurement of chromatic dispersion of polarization modes in optical fibres using white-light spectral interferometry," *Meas. Sci. Technol.*, vol. 21, Mar. 2010. Accessed: Nov. 27, 2025. DOI: 10.1088/0957-0233/21/4/045302.
- [6] S. Dwivedi, *et al.*, "Experimental Extraction of Effective Refractive Index and Thermo-Optic Coefficients of Silicon-on-Insulator Waveguides Using Interferometers," *Journal of Lightwave Technology*, vol. 33, 2015. Available: <https://ieeexplore.ieee.org/document/7243301>.
- [7] X. Chen, J. Lin, and K. Wang, "A Review of Silicon-Based Integrated Optical Switches," *Laser & Photonics Reviews*, vol.17, 2023. Available: <https://onlinelibrary.wiley.com/doi/full/10.1002/lpor.202200571>.
- [8] Applied Nanotools, "NanoSOI Fabrication Service." www.appliednt.com. <https://www.appliednt.com/nanosoi-fabrication-service/> (accessed Dec. 1, 2025).
- [9] K. Okamoto, *Fundamentals of Optical Waveguides*. Elsevier: United Kingdom, 2022, pp.154.